

Yukta Seelam

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SUMMARY

Graduate student in Robotics and Autonomous Systems at Boston University with hands-on experience in reinforcement learning, perception, and robot control. Skilled in designing and integrating full-stack autonomous systems using ROS, CARLA, and computer vision pipelines for real-world robotics applications.

ACADEMIC PROFILE

Boston University **3.8/4.0**
Masters in Robotics and Autonomous Systems

Expected Jan 2026

Indian Institute of Technology Kanpur
B. Tech in Electrical Engineering

May 2024

EXPERIENCE & RESEARCH

Hybrid Local-Cloud RL for Autonomous Driving in CARLA – Red Hat / Boston University

May 2025 – Sep 2025

- Developed and trained a hybrid local-cloud reinforcement learning policy (UniLCD) in the CARLA simulator, optimizing for latency, safety, and energy efficiency in autonomous driving scenarios.
- Resolved headless rendering issues and implemented a video logging pipeline by using OpenCV backends, enabling visualization of training and evaluation rollouts.
- Conducted large-scale training with checkpoint recovery, implemented custom progress logging, and performed policy evaluation using RGB, depth, and segmentation sensor data.

Obstacle Avoidance and Path Planning for Rover – IIT Kanpur

Aug 2023 – Mar 2024

- Developed perception-based obstacle avoidance algorithms for the Clearpath Husky Rover in dynamic environments.
- Fine-tuned YOLOv8 models for fruit-harvesting tasks, improving detection sensitivity and navigation accuracy.
- Integrated ROS-based A* and TEB planners for real-time autonomous path generation.

Samsung Research & Development – India

May 2023 – Jul 2023

- Implemented CNN-based object detection (YOLOv5, RefineDet) for robotic vacuum cleaners to improve real-world navigation accuracy.
- Collaborated with robotics and ML teams to deploy models for cross-platform compatibility.
- Automated IP whitelisting using AWS Lambda and SNS, enhancing system security and operational efficiency.

Development of AI Combat Robots – IIT Kanpur / IIT Mandi

Dec 2021 – May 2022

- Built two autonomous combat robots achieving 85% mAP for real-time enemy detection and 90% accuracy in dynamic target shooting.
- Performed advanced localization and path planning for combat robots in dynamic arenas using visual markers, Eigen transforms, A* planning, and embedded NVIDIA GPUs, optimizing for high perception and SLAM rates.

AWARDS AND PUBLICATIONS

- 3rd Place – IEEE ICRA DJI RoboMaster AI Challenge 2022 (only Indian team internationally)
- Paper: “Towards Autonomous Shooting Rover via Situation-Aware Visual Perception and Dynamic Action Execution”, IEEE/SICE SII 2024 (Vietnam)
- Poster: “Design of AI Ground Robots for Intelligent Combat in a Dynamic Arena”, ICRA 2022

LEADERSHIP POSITIONS

- Team Head – Robotics Club - IITK | President – Toastmasters IIT Kanpur | Captain – IITK Basketball Team | Student Ambassador – BU Robotics

SKILLS & COURSEWORK

- Programming: Python, C++, MATLAB, Bash, SQL
- Robotics Tools: ROS, Gazebo, RViz, CARLA, OpenCV, SLAM, Control Systems, Path Planning
- ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas
- Hardware & Design: Fusion 360, SolidWorks, Eagle, Arduino, Embedded Systems